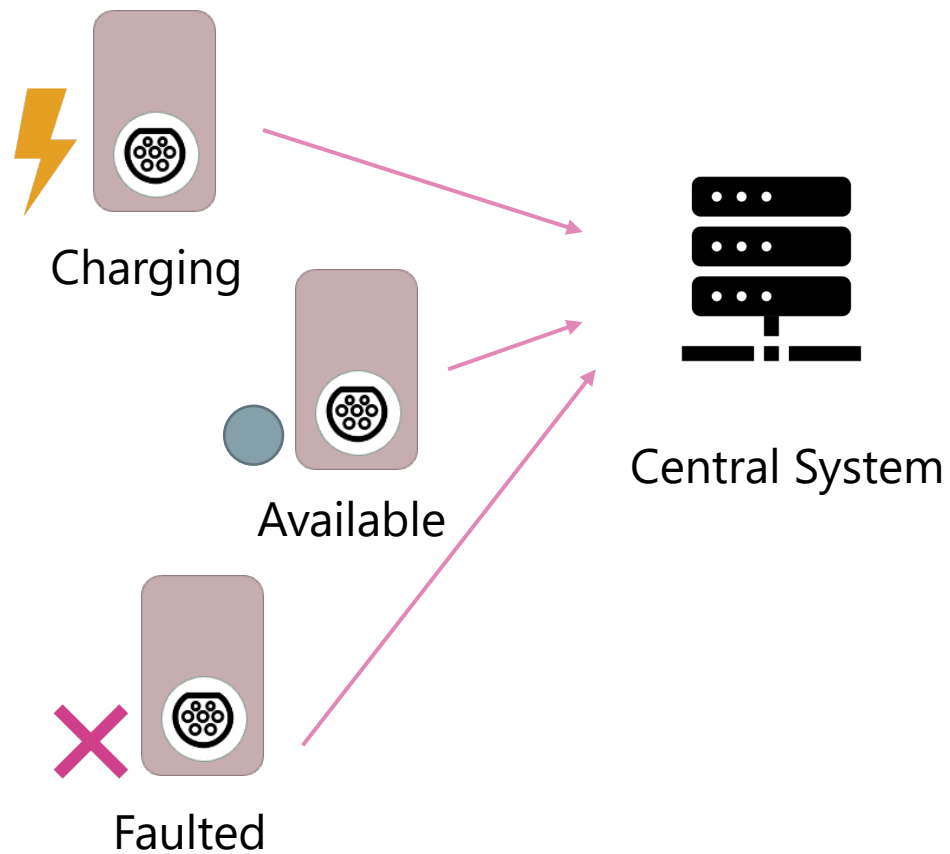


4. CHARGING STATION STATUSES



Charging Station Status



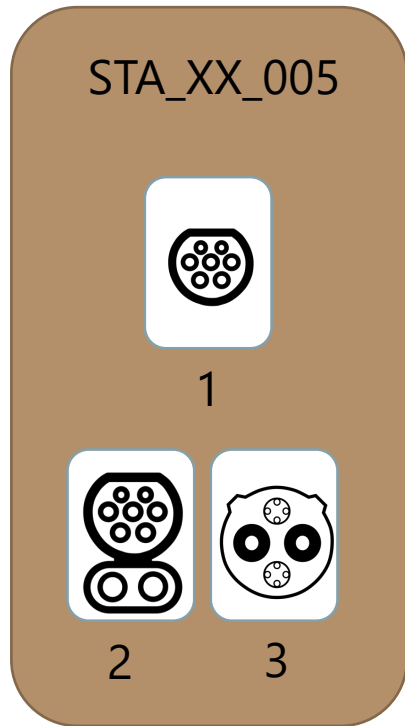
Charging stations can have different **operational statuses**. A status is "**Available**" when a station is free and nobody is using it, "**Charging**" when there is an EV charging at the station, or "**Faulted**" if there is some error on the station.

A status is always **linked to a single connector** of a station. For example, if a station has two connectors, one may be "Available," while another is "Charging."

Stations report statuses to the central system with a message called "**Status Notification.**"



Status of Different Connectors



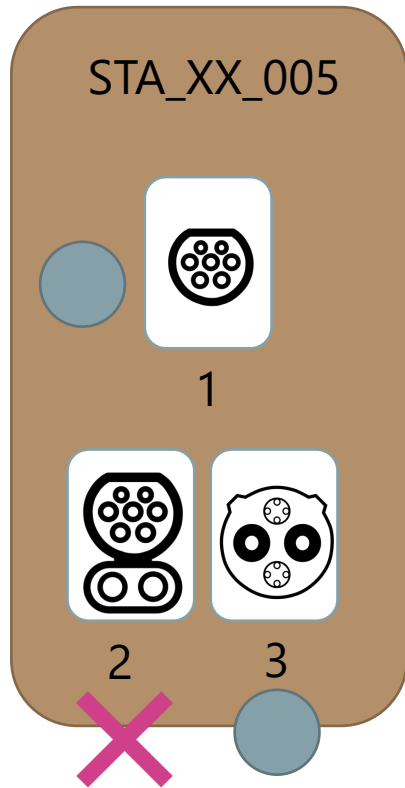
As we remember, a charging station may have **multiple connectors**. Each connector has a unique number called a **connector ID**. The status of each connector is linked to its connector ID.

For example, if a station starts charging on the CCS connector (with connector ID 2), it might report the following status to the central system

- **Connector 1** (Type 2): status = **Available**
- **Connector 2** (CCS): status = **Charging**
- **Connector 3** (ChaDeMo): status = **Unavailable**



Error reporting



Each connector has a **status** (available, occupied, etc.) and **error code**. If a connector does not have an error, the error code is NoError.

If a connector becomes faulted for some reason, the station reports to the Central System which connectors are faulted and what the fault is. It may also report a vendor-specific, more detailed error code.

Example: If the **CSS connector (2)** of a station **becomes faulted**, it could report the follow status to the Central System

- Connector = **2**, status = **Faulted**, error code = **OtherError**, vendor error code = **gf_393.3**



Error Codes

The OCPP standard defines around 20 different **standard error codes** that stations may report. In addition to these standard error codes, stations may have their own custom error codes. For example, a station may send an errorCode of OtherError and a vendorErrorCode of err_xy_112, where OtherError is one of OCPP's standard error codes and err_xy_112 is a vendor-specific custom error code.

Few of the most common standard error codes.

Status	Description
NoError	This error code is reported when a station reports a new status but does not have any error. For example, when an electric vehicle (EV) starts charging, the station might report a status of Charging and an errorCode of NoError.
ConnectorLockFailure	This is a fairly common error that can occur when a driver tries to start charging but the charging cable is not fully inserted into the socket and the station is not able to lock the cable.
OtherError	An error situation that does not fit any of OCPP's standard error codes can be reported using the OtherError code. In this case, more details about the error can be provided in the vendorErrorCode field.
WeakSignal	The WeakSignal error code is often used to indicate that wireless communication is not functioning properly. A typical situation in which this error might occur is an underground garage with a weak GSM signal, resulting in the station's modem not being able to maintain a reliable connection.



StatusNotification.req message, required fields

Example:

```
[2, "533-aa-22911" ,  
  "StatusNotification",  
  {  
    "connectorId":1,  
    "errorCode":"NoError",  
    "status":"Available"  
  }  
]
```

Each status notification that a station sends to the Central System must contain at least the following three required fields:

1. **Connector ID:** This field specifies **which connector** the status is related to. If the status changes on multiple connectors, the station should send separate status notification messages for each connector.
2. **Error Code:** Status notifications always have an error code. If there is no error, the error code is NoError.
3. **Status:** This field specifies the new status of the connector that has changed. Typical statuses include Available, Unavailable, etc.



Different status types

Status	Description
Available	A connector is working normally and an EV can start using it
Preparing	Usage of a connector is initiated, meaning an EV will start charging soon. But the actual charging (= providing energy from station to the EV) has not started yet.
Charging	An EV is charging at the station
SuspendedEV	The station would be ready to provide electricity to the EV (the same way as in Charging status), but the EV has suspended the charging and does not take in energy at the moment. The same charging transaction will continue later when the EV again decides to take in energy.
SuspendedEVSE	Station and EV are in charging mode, but the EVSE (meaning station) has suspended providing energy at the moment. The same charging transaction will continue later when the station again decides to provide energy to the EV.
Finishing	Stopping the charging has been initiated, but the actual charging transaction has not yet been stopped. It may for example be waiting for the EV driver to disconnect the cable.
Reserved	A charging station (or station's connector) has been reserved for an EV driver, but that EV driver has not yet arrived to the station and started the charging.
Unavailable	A connector cannot be used at the moment. A typical situation could be that only on DC connector, CCS or ChaDeMo, can be used at the same time. So if an EV is using CCS (Charging), then ChaDeMo cannot be used at the same time (Unavailable)
Faulted	A connector has a fault and it cannot be used at the moment.



StatusNotification.req message, optional fields

Stations may report additional details, especially in error situations, about the status of the various connectors on the station. This optional information may include:

Field	Card.	Description
info	0..1	A free text field providing more information about the error. This field has a maximum length of 50 characters.
timestamp	0..1	A timestamp indicating when the status changed on the station. For example, if a station reports a fault, this timestamp should be the time when the fault started. Please note that this may not be the same time as when the message was sent and received if there is a delay in sending by the station or receiving by the central system.
vendorId	0..1	A field identifying the vendor if there is vendor-specific implementation, such as vendor-specific error codes or messages. This is often the same as the vendor reported in the boot notification, but it could potentially be a different ID.
vendorErrorCode	0..1	A vendor-specific error code field. Please note that this only includes the error code itself (e.g., "err_xyz_123") and does not provide a human-readable explanation of the code. Charging station vendors may provide documentation on different error codes through their own processes, but this is outside the scope of OCPP.



StatusNotification.conf message

Example:

```
[3, "533-aa-22911", {}]
```

The Central System's response to a station's status notification message is a simple, empty message acknowledging receipt of the message.

